

AI/ML TASK 2 – MODEL COMPARISON
REPORT

Feature Engineering, Model Optimization & Performance Comparison

House Price Prediction using the California Housing Dataset
Dataset

METHODOLOGY

Approach Overview

- Dataset: California Housing (sklearn.datasets)
- Preprocessing: Train/Test split 80/20, StandardScaler fitted on training data only
- Models trained: Linear Regression, Ridge Regression, Decision Tree, Random Forest, Gradient Boosting, Support Vector Regression
- Evaluation metrics: MAE, MSE, RMSE, R^2

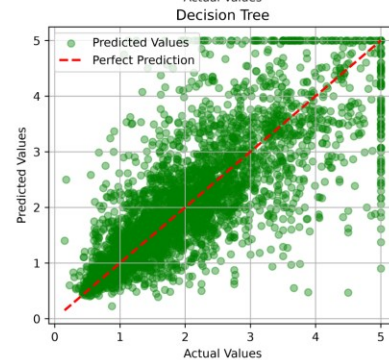
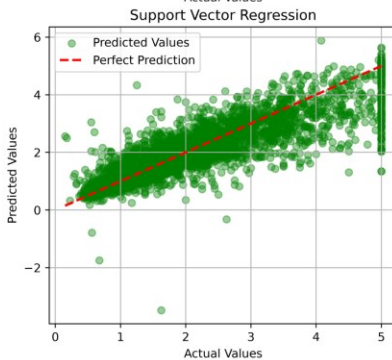
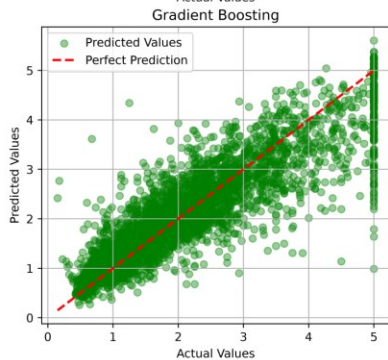
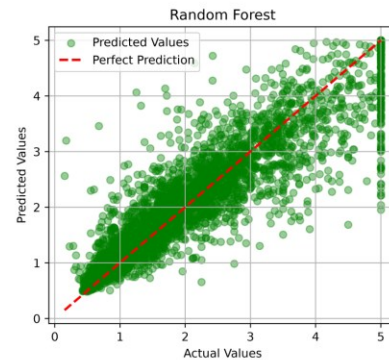
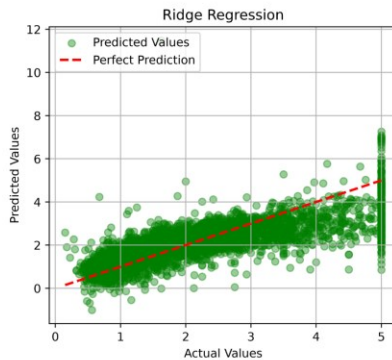
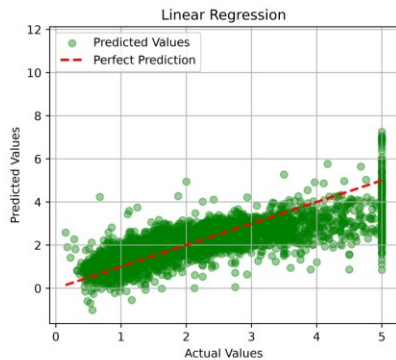
RESULTS

Model Performance Comparison Comparison

Model	MAE	MSE	RMSE	R ² Score
Random Forest	0.3278	0.2559	0.5059	0.8047
Gradient Boosting	0.3717	0.2940	0.5422	0.7756
Support Vector Regression	0.3738	0.3245	0.5697	0.7524
Decision Tree	0.4532	0.4930	0.7021	0.6238
Linear Regression	0.5332	0.5559	0.7456	0.5758
Ridge Regression	0.5332	0.5559	0.7456	0.5758

VISUAL COMPARISON

Actual vs Predicted Scatter Plot Grid



CONCLUSION

Random Forest Selected as Final Model

Lowest error metrics and highest R^2 (0.8047).
Ensemble methods outperform linear and single-tree
single-tree models on this dataset.